

COLESHILL, United Kingdom

WMO: 035350

Lat: 52.4800N

Lon: 1.6908W

Elev: 96

StdP: 100.18

Time Zone: 0.00 (EUW)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.9	-2.4	-5.7	2.4	-1.9	-4.4	2.6	-0.9	15.6	8.8	13.1	9.4	2.0	350	0.567

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	26.4	18.7	24.3	17.7	22.5	16.8	19.6	24.7	18.5	22.7	17.6	21.3	3.5	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.7	12.9	21.7	16.8	12.1	20.3	16.0	11.5	19.3	56.3	24.7	52.7	22.7	49.7	21.4	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.3	9.2	7.8	DB	-6.3	30.0	1.8	2.5	-7.6	31.7	-8.7	33.2	-9.7	34.6	-11.0	36.3	(4)
(5)			WB	-6.5	22.0	1.8	1.6	-7.8	23.1	-8.9	24.1	-9.9	24.9	-11.2	26.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	4.5	4.8	6.5	8.9	11.7	14.9	16.8	16.4	14.1	11.0	7.2	5.0
	DBStd	5.32	3.28	3.33	3.05	2.90	2.77	2.54	2.60	2.29	2.48	2.97	3.14	3.61
	HDD10.0	774	170	149	114	53	14	1	0	0	1	24	92	157
	HDD18.3	3009	427	380	368	283	207	108	63	69	128	229	334	413
	CDD10.0	838	1	2	4	21	65	147	212	198	125	53	8	3
	CDD18.3	31	0	0	0	0	0	4	16	9	2	1	0	0
	CDH23.3	281	0	0	0	1	7	47	154	57	12	3	0	0
	CDH26.7	49	0	0	0	0	0	6	32	10	1	1	0	0
Wind	WSAvg	3.6	4.3	4.2	4.1	3.8	3.7	3.2	3.0	3.0	3.0	3.6	3.7	3.9
Precipitation	PrecAvg	672	64	47	49	48	54	57	50	65	58	54	61	67
	PrecMax	912	131	139	89	105	170	146	109	120	138	125	133	138
	PrecMin	492	14	8	5	2	13	9	4	9	11	5	25	12
	PrecStd	91	29	29	23	25	34	34	27	29	36	30	25	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.2	14.1	16.7	21.8	24.3	27.6	29.9	27.7	25.1	20.1	15.4	13.7
		MCWB	11.4	11.6	11.2	14.5	17.3	19.4	19.9	19.4	17.6	16.3	13.7	12.0
	2%	DB	11.7	12.1	14.2	18.3	21.7	24.6	27.0	24.6	22.2	17.7	14.0	12.5
		MCWB	10.3	10.2	10.4	12.9	15.8	18.0	18.8	18.2	17.1	15.0	12.5	11.2
	5%	DB	10.4	10.9	12.8	16.1	19.2	22.4	24.9	22.6	20.4	16.4	13.0	11.4
		MCWB	9.2	9.2	9.8	11.6	14.3	16.7	17.9	17.2	16.0	14.2	11.6	10.3
	10%	DB	9.4	9.7	11.5	14.2	17.0	20.4	22.5	21.1	18.8	15.3	11.9	10.3
		MCWB	8.1	8.2	9.0	10.5	12.8	15.8	16.8	16.3	15.1	13.4	10.7	9.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.7	11.8	12.2	15.4	18.1	21.3	21.2	20.5	19.5	16.8	14.2	12.4
		MCDB	12.7	13.1	14.6	20.8	22.4	26.3	27.7	25.9	23.1	19.3	15.0	13.2
	2%	WB	10.5	10.5	11.1	13.5	16.3	18.8	19.6	19.1	17.8	15.7	12.8	11.4
		MCDB	11.4	11.9	13.2	17.3	20.9	23.1	25.4	23.2	20.9	17.3	13.7	12.2
	5%	WB	9.4	9.5	10.2	12.1	14.9	17.3	18.4	18.1	16.7	14.7	11.8	10.4
		MCDB	10.3	10.7	12.4	15.3	18.3	21.2	23.2	21.6	19.1	16.0	12.8	11.3
	10%	WB	8.3	8.4	9.2	11.0	13.6	16.2	17.4	17.1	15.7	13.7	10.7	9.2
		MCDB	9.3	9.6	11.3	14.0	16.3	19.6	21.6	20.0	18.0	15.0	11.7	10.1
Mean Daily Temperature Range	5% DB	MDBR	5.2	5.7	7.3	8.7	8.8	9.0	9.3	8.7	8.6	6.8	5.6	5.1
		MCDBR	6.1	6.6	9.3	11.9	12.2	12.8	13.3	11.1	10.7	7.5	6.2	5.6
		MCWBR	5.7	5.1	6.2	7.2	7.1	7.1	6.1	5.6	6.0	5.1	5.3	5.2
	5% WB	MCDBR	6.1	6.3	8.6	10.8	10.7	11.0	11.5	9.8	9.2	6.9	6.0	5.6
		MCWBR	5.8	5.2	6.1	6.9	6.6	6.9	6.1	5.5	6.0	5.3	5.5	5.4
Clear-Sky Solar Irradiance	taub	0.321	0.337	0.369	0.394	0.399	0.397	0.396	0.382	0.366	0.349	0.324	0.311	
	taud	2.360	2.390	2.297	2.235	2.260	2.292	2.299	2.361	2.407	2.435	2.411	2.337	
	Ebn at Noon	666	773	814	837	851	856	849	842	812	749	661	605	
	Edh at Noon	61	78	103	124	127	124	122	108	92	74	57	52	
All-Sky Solar Radiation	RadAvg	0.75	1.44	2.52	3.86	4.77	5.10	4.88	4.02	3.02	1.75	0.92	0.61	
	RadStd	0.07	0.17	0.30	0.42	0.30	0.51	0.53	0.29	0.27	0.15	0.10	0.06	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (93 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page